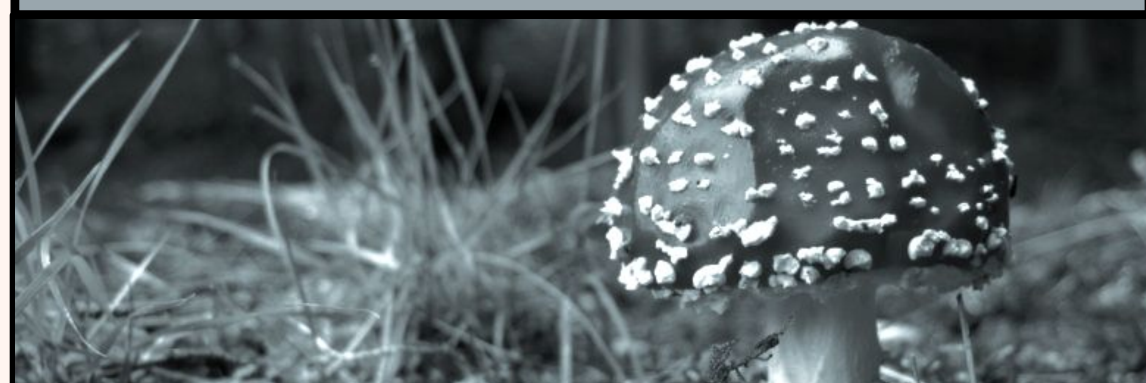
Which Mushrooms Can We Eat?

A decision tree rule of thumb for eating mushrooms

2023

If you are in a real life survival situation, please do not choose to eat mushrooms unless you are qualified to accurately identify them.

Introduction



Abstract

The goal of this study was to find a good rule of thumb for consuming mushrooms in a survival situation. The rule of thumb was obtained using a tree model fitted on one portion of the data and tested on another portion. This gives a good approximation of the true power of the classifier.

The Dataset

This dataset, provided by the UCI Berkeley Machine Learning Repository, is called the mushroom dataset. It contains 23 features including the following:

- Classification of edible or poisonous
- Physical traits
- Odor traits
- Species information

Research

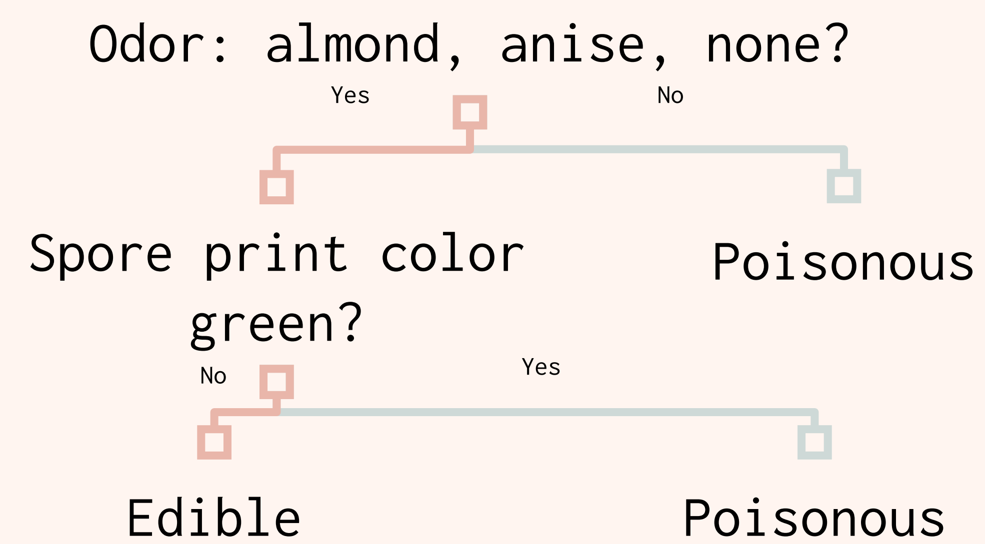
Methodology

1. Use a decision tree model to classify the mushrooms
2. Measure the proportion of misclassified mushrooms
3. Show that no rule of thumb can be used for the remaining mushrooms

	A training set is selected from the data to fit the tree
	The tree is tested on the test set to approximate its accuracy

Analysis

By training a tree model on the dataset, we find that the most accurate decision tree only has three leaf nodes and five total nodes.



The decision is largely based on the odor of the mushroom, but also considers the spore print color.

Poisonous (in test set)	Edible (in test set)
1526	1598

Unfortunately, in order to get a spore print, it takes 2-24 hours for the print to develop. In a survival situation, this may not be possible. When we exclude spore print colors, the best model we can find uses only the odor to classify.

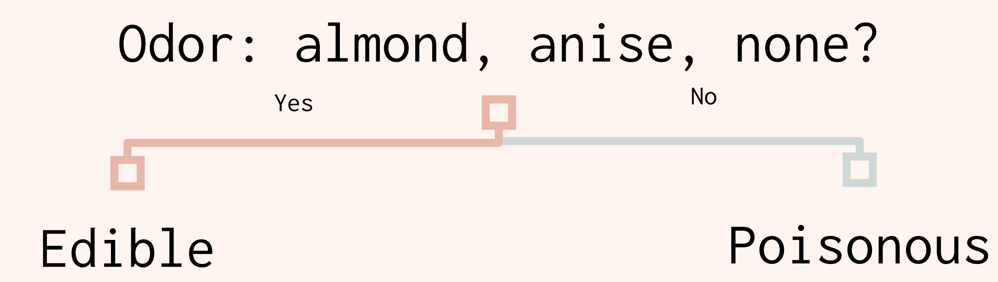
Classifier Status	Poisonous	Edible
Correct	1510	1598
Incorrect	16	0
Simple Correct	1482	1598
Simple Incorrect	44	0

The training set is 5000 data entries long and the test set is 3124 entries long. The simple classifier is 98-98.96% accurate at the 95% confidence level, and the more complex classifier is 99.1-99.7% accurate at the 95% confidence level.

Conclusion

Conclusion

As found in this analysis, there is not a 100% accurate decision tree for classifying mushrooms as poisonous or edible. Although a classifier that only misclassified edible mushrooms was attempted, it caused too much of an accuracy decrease to be useful.



Recommendations

In general, do not choose to eat wild mushrooms unless you know what you are doing. This classifier is definitely not safe for the general public to use regularly. The likelihood that you select non-poisonous mushrooms even 20 times consecutively is approximately 73.8%, which is extremely low considering the repercussions of even one wrong choice. Use this information at your own risk.

Sources

Dataset: UCI Berkeley Machine Learning Repository
Mushroom Print Facts:
https://namyco.org/interests/education/how-to-make-a-spore-print/

Author

William Bradford
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